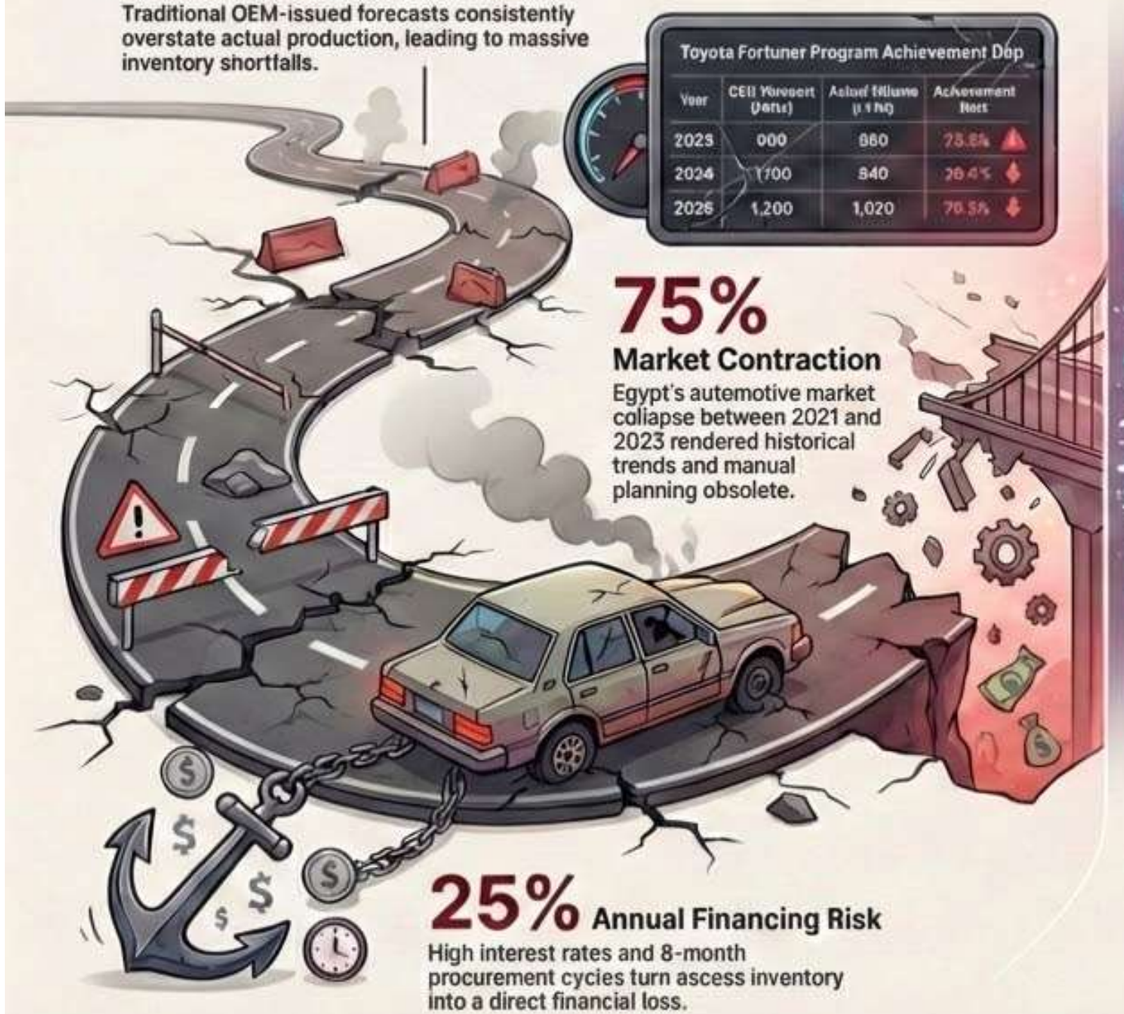
Supply Chain Distribution Prediction Using AI

The Cost of Traditional Planning Gaps

24.3% Average Forecast Error

Traditional OEM-issued forecasts consistently overstate actual production, leading to massive inventory shortfalls.



75%

Market Contraction

Egypt's autemotive market collapse between 2021 and 2023 rendered historical trends and manual planning obsolete.

25% Annual Financing Risk

High interest rates and 8-month procurement cycles turn ascess inventory into a direct financial loss.

The AI-Powered Predictive Solution

88%

Predictive Correlation

External AMIC market data explains nearly all variance in program achievement rates.

LSTM

Deep Learning Integration

Using Long Short-Term Memory networks to capture complex patterns in FX rates and production.

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Deep Learning Integration

Using Long Short-Term Memory networks to capture complex patterns in FX rates and production.

37.1%

Faster Disruption Response

AI-driven systems significantly outperform traditional manual planning in reaction speed and risk mitigation.

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Decision Support System (DSS)

An information system supporting organizational decisions, designed to shift Managers from "doers" to **Strategic Auditors**.

K-Means Clustering

An unsupervised algorithm grouping data points into K clusters. Strategically used to identify **Market Regimes** and hidden patterns in logistics and demand.

Built on SDLC (Software Development Life Cycle)

The entire architecture is built using structured **Meccano (Modular) Logic**, ensuring 100% reusability and rapid AI configuration across the pipeline.

Scenario Generator

Prescriptive Alert Engine

Supply Chain Manager Dashboard

Unsupervised K-Means Clustering (Market Regimes)

Supervised LSTM Neural Network (Time-series) & Random Forest

RPA Data Ingestion Bots

LSTM & Random Forest

LSTM: A specialized Recurrent Neural Network (RNN) capturing long-term dependencies, essential for **Time-series forecasting** in production.

Random Forest: An ensemble method using multiple decision trees for highly accurate classifications and regressions of complex supply chain variables.

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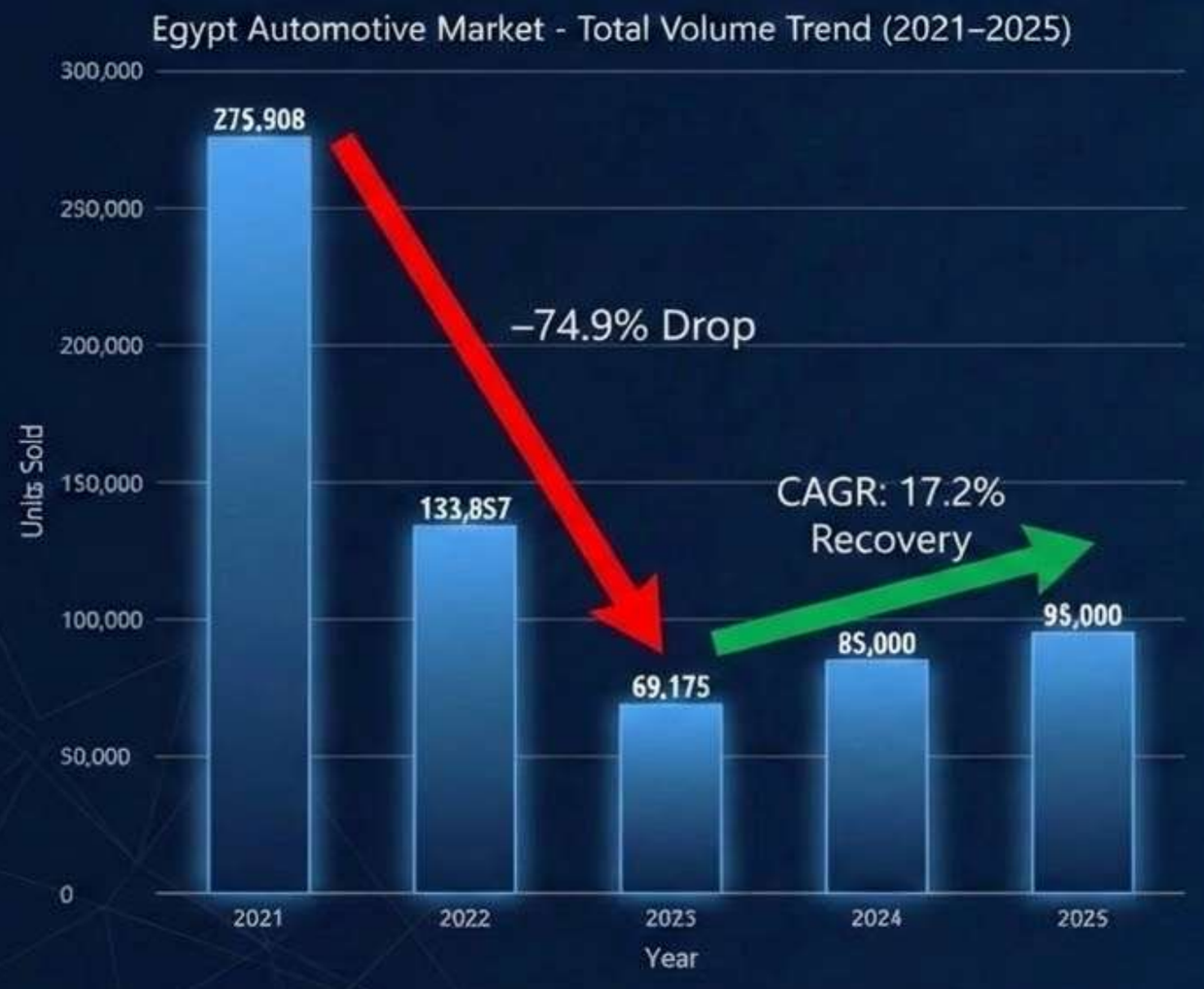
Egypt Automotive Market – Growth Trend (2021–2025)

AMIC Analysis | Mobica Automotive



Executive Summary

- Peak Volume (2021): **275,908 units**
- Trough (2023): **69,175 units**
- Peak-to-Trough Decline: **–74.9%**
- Peak-to-Trough Decline: **–74.9%**
- Recovery (2025 Est.): **95,000 units**
- CAGR (2023—2025): **17.2%**

Key Drivers

- FX Crisis: EGP devaluation drove **–75%** demand collapse
- Recovery: **+22.9%** (2024), **+11.8%** (2025 est.) as restrictions eased
- AI-DSS: LSTM forecasting cut Fortuner MAPE **27.8% → 21.5%**
- Outlook: Full recovery to **–275K** not expected before 2028—2030

Macroeconomic Shocks Invalidate Static Planning Models

Achievement Rate (%) =
0.000228 × AMIC Sales + 57.26

R^2 = 0.88. Public AMIC data explains 88% of the variance in Toyota Fortuner achievement rates. It is the missing predictive key.

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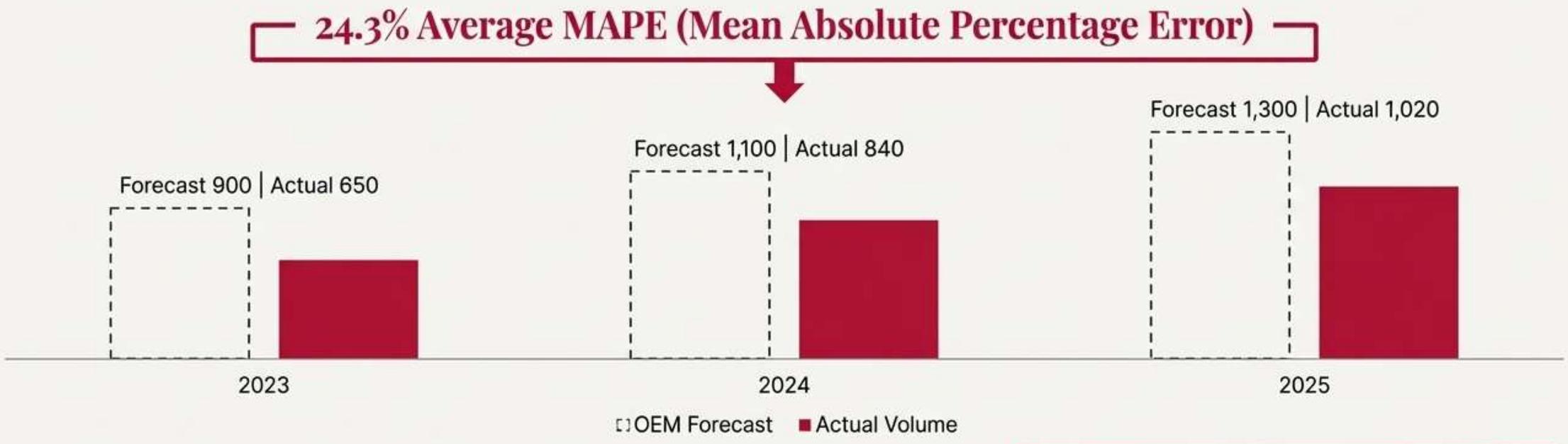
Premium, Print-Ready Executive Briefing Slide

The Structural Failure of OEM Forecasting

Toyota Fortuner Program Volume Assessment (2023–2025)

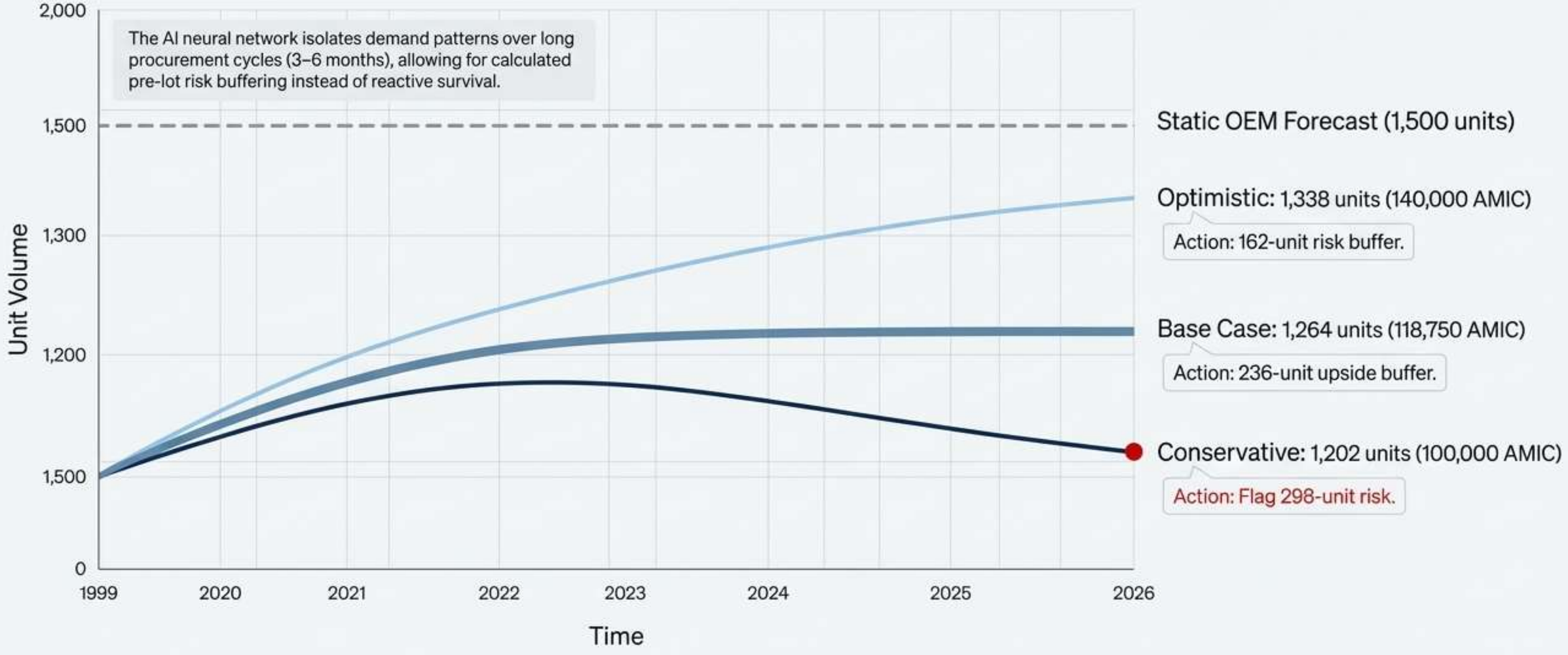


Mobica operates entirely blind, reliant on single-source volume letters that systematically overstate market reality.

Critical Failure Sub-Case: GM C43 Program

OEM Forecast: 1,248 units | **Actual Delivered:** 276 units | **Result:** A devastating 22.1% achievement rate, confirming multi-program systemic failure.

LSTM Provides Probabilistic Boundaries, Not Single Points of Failure



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Translating Algorithmic Output into Prescriptive Action

Procurement Strategy

- Plan TTAP import orders for 1,264 units (Base Case), overriding the OEM 1,500 forecast.
- **Impact: Eliminates EGP 247,800 in annual excess financing.**

Pricing Strategy

- Negotiate a 3–5% volume risk premium for Round 5 pricing.
- Structure tiered bands based on historic AI-proven variance (<65%, 65-80%, >80%).

Risk Management

- Implement a 120-day pre-lot TTAP checkpoint.
- **Action:** Delay order by one cycle if AI predicts <75% achievement.

ERP Integration

- Feed DSS predictions directly into MRP planning parameters.
- **Outcome:** AI actively adjusts the system demand signal without manual data entry.

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From Reactive Survival to Proactive Optimization

Metric	Pre-AI Baseline	Post-AI Target
Forecast Accuracy (MAPE)	24.3% (3-yr avg)	**< 10%**
Procurement Alert Response	30+ days lag (reactive)	**< 5 days** (proactive)
Excess Inventory Ratio	~24% of annual volume	**< 10%**
AI Model Update Cycle	Manual / Non-existent	Monthly (Automated RPA)
Finance Cost Savings	EGP 0	**EGP 247,800+ annually**

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Reengineering the Planning Workflow

Traditional Workflow

OEM annual volume letter

↓

Supply Chain manual data entry

↓

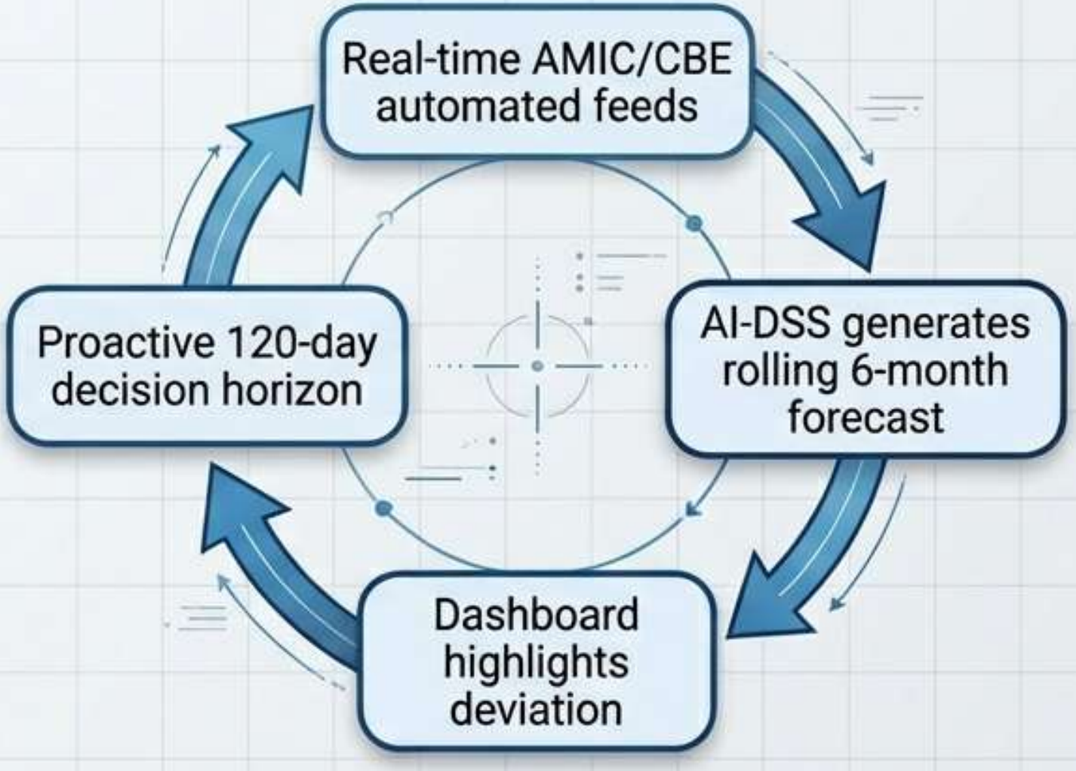
Blind TTAP orders

↓

30-day lag reactive adjustments

Mindset: 100% Deference to OEM forecasts. Manual Excel snapshots.

AI-Driven Workflow



Mindset: Data-supported counter-forecasts. Evidence-based negotiation.

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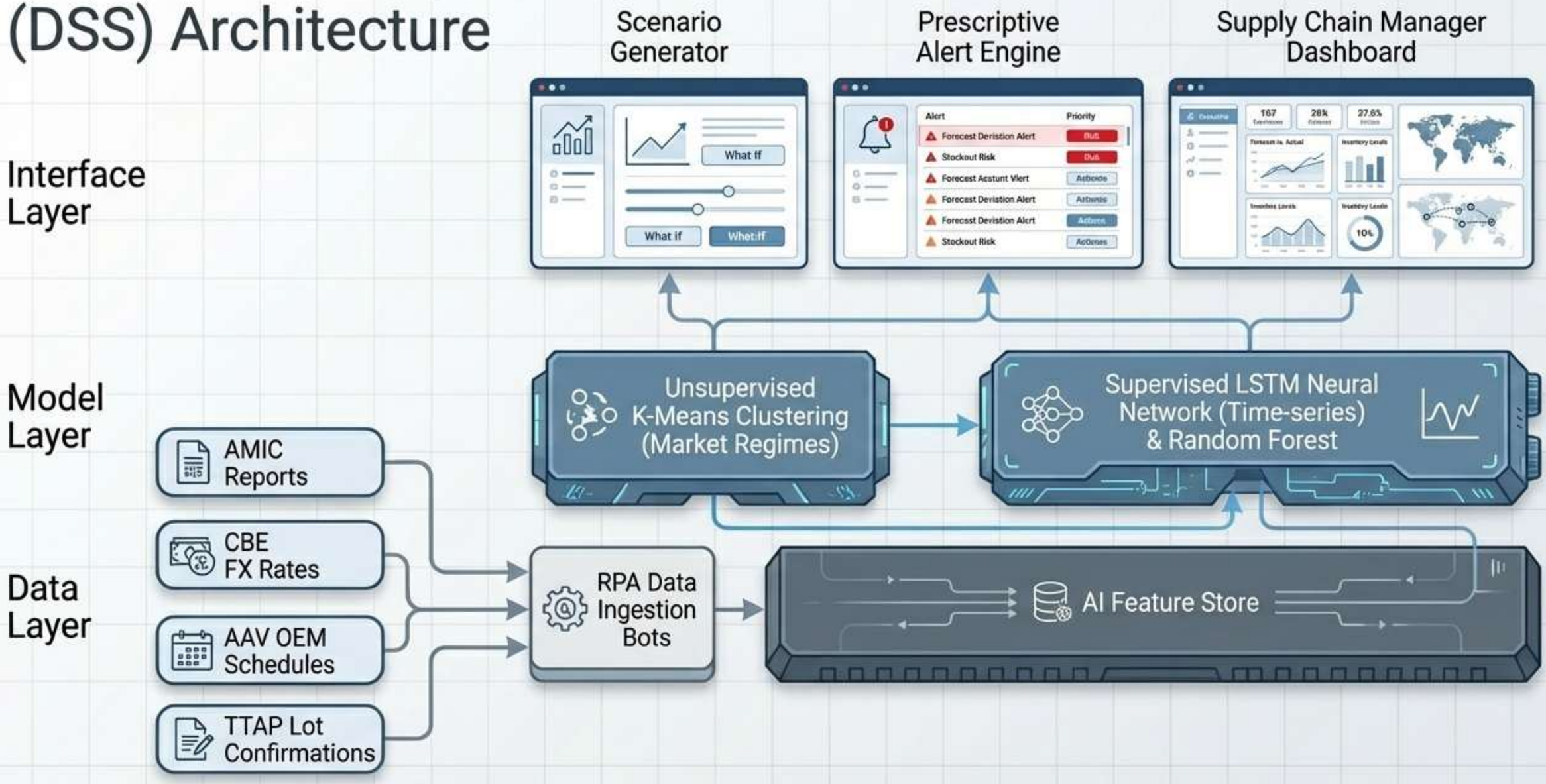
The Decision Support System (DSS) Architecture



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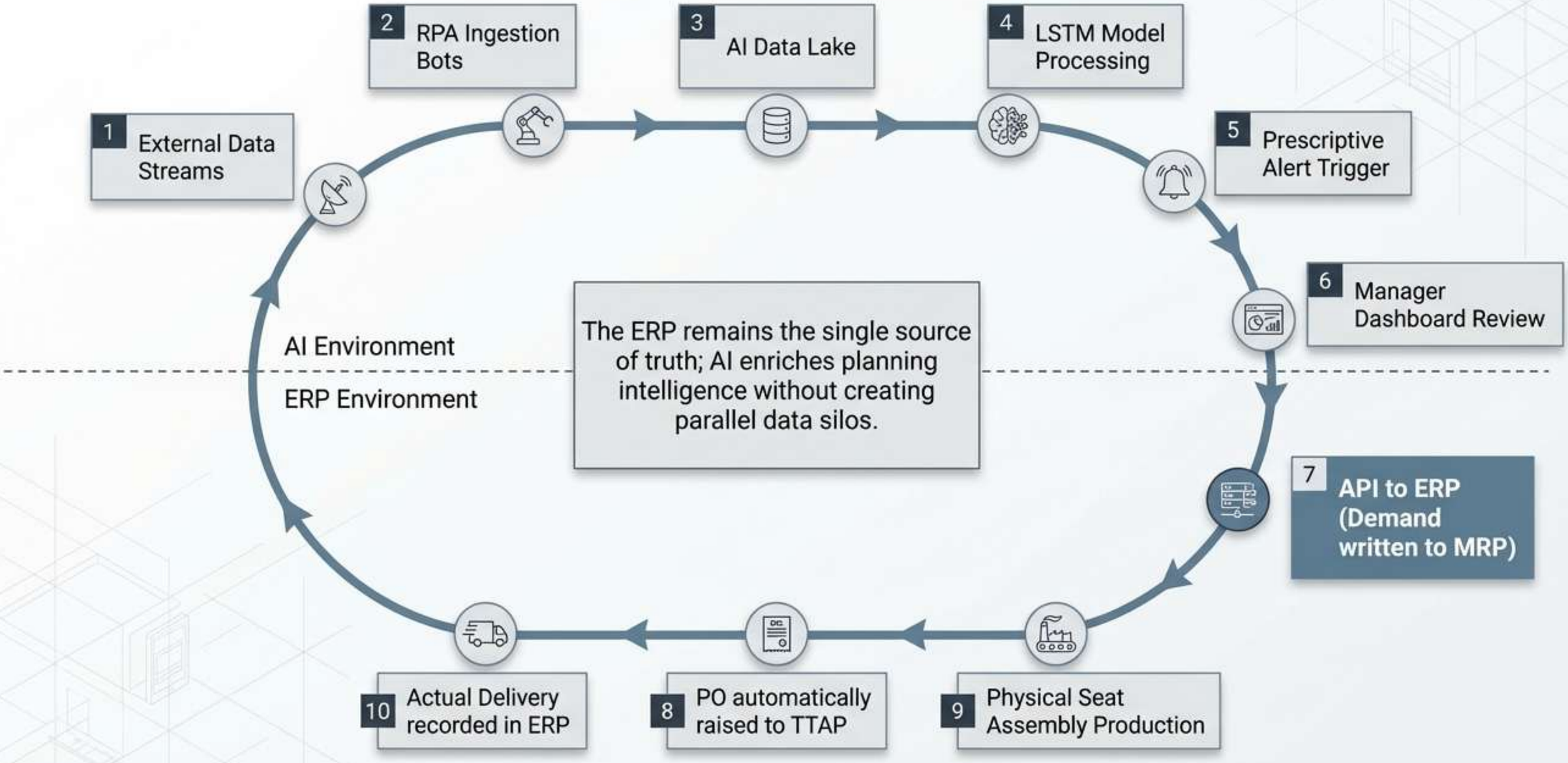
Seamless ERP Integration: A Closed-Loop Topology



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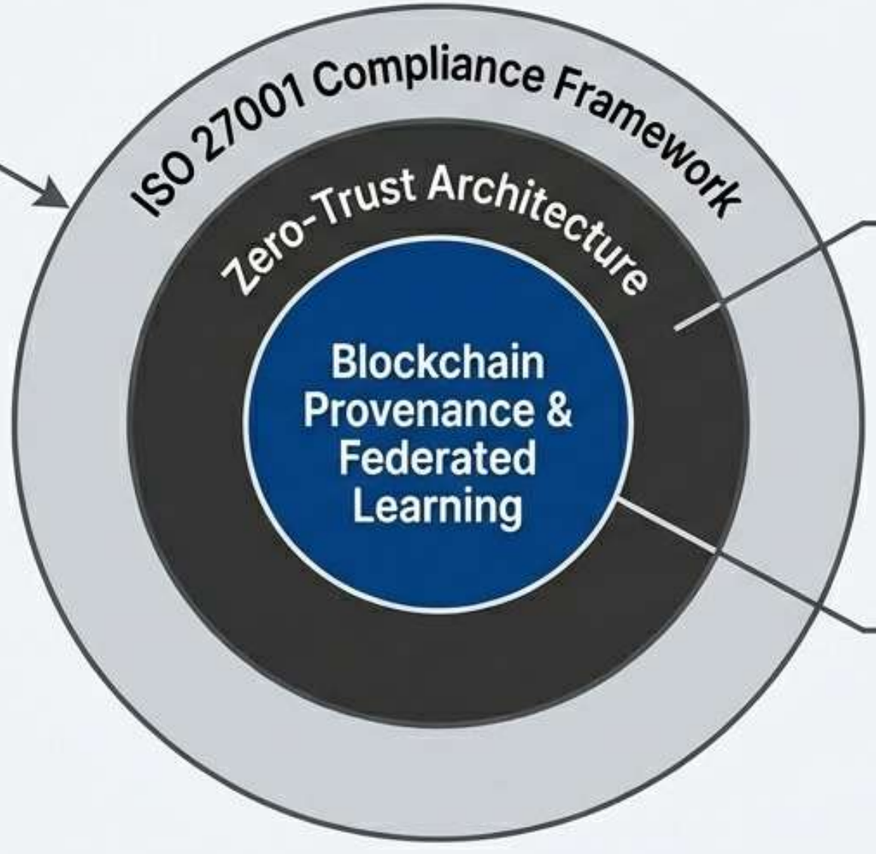
Premium executive briefing slide

Securing the Supply Chain Data Supply

Ensuring global standard adherence for information security across all external API connections.



Strict, continuous identity verification for all internal queries and external data calls.

Ensuring highly sensitive OEM pricing and AAV scheduling data cannot be poisoned, stolen, or manipulated.

Implementing data governance that fundamentally **protects** sensitive Tier-1 commercial secrets while

Implementation Roadmap: 18 Weeks to Go-Live

Week 1 | Week 4 | Week 5 | Week 10 | Week 11 | Week 14 | Week 15 | Week 18

Sprint 1: Foundation (Weeks 1-4)

Data architecture, AMIC API connection, ERP data extract, and RPA bot development.

Sprint 2: Model Development (Weeks 5-10)

Feature engineering, LSTM baseline training (2023-2025 data), and backtesting against baseline MAPE.

Sprint 3: DSS Interface (Weeks 11-14)

UI Dashboard MVP, scenario generator, and RPA alert integration.

Sprint 4: Deployment (Weeks 15-18)

User Acceptance Testing (UAT), Supply Chain team training, go-live, and performance monitoring setup.

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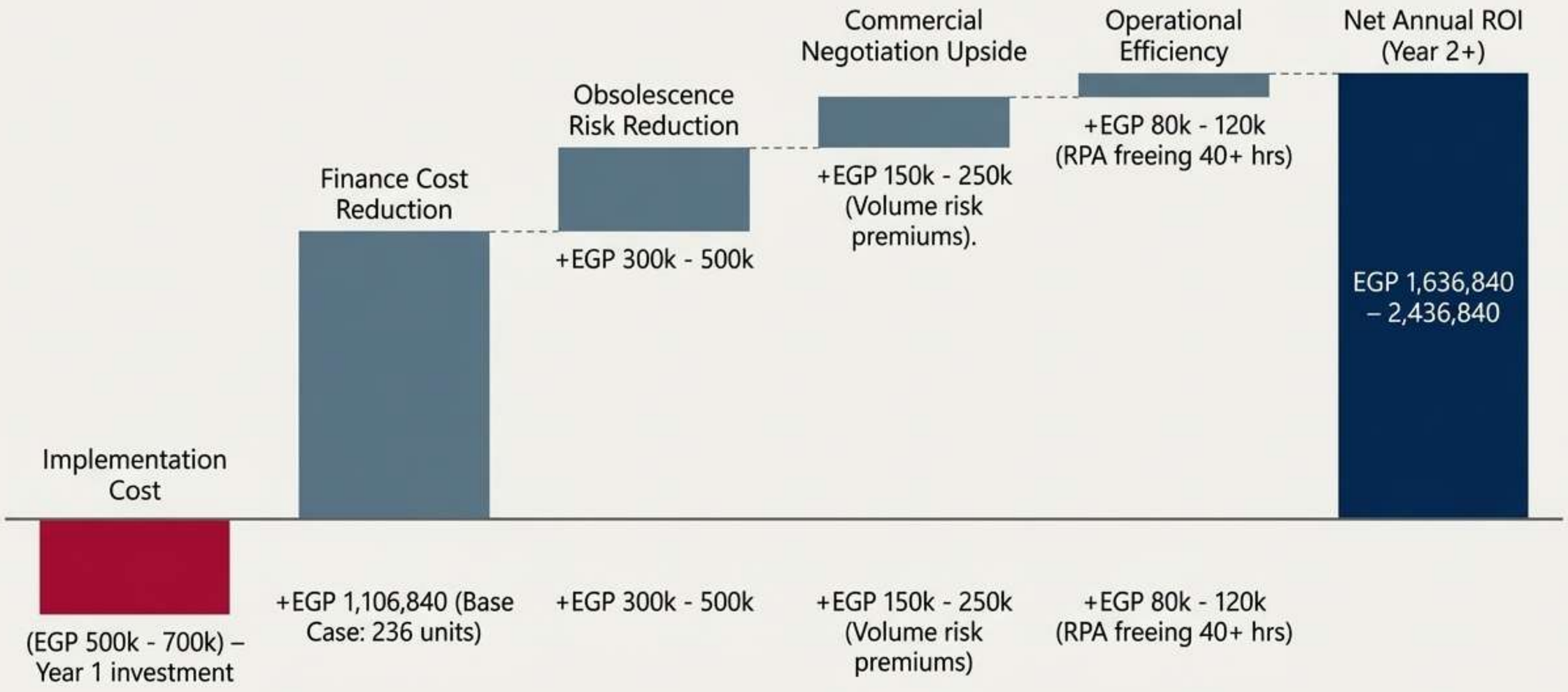
Financial Impact: Self-Funding by Year 2

Key Insight: Self-funding initiative with net positive ROI achieved in Year 1, scaling exponentially in Year 2+.



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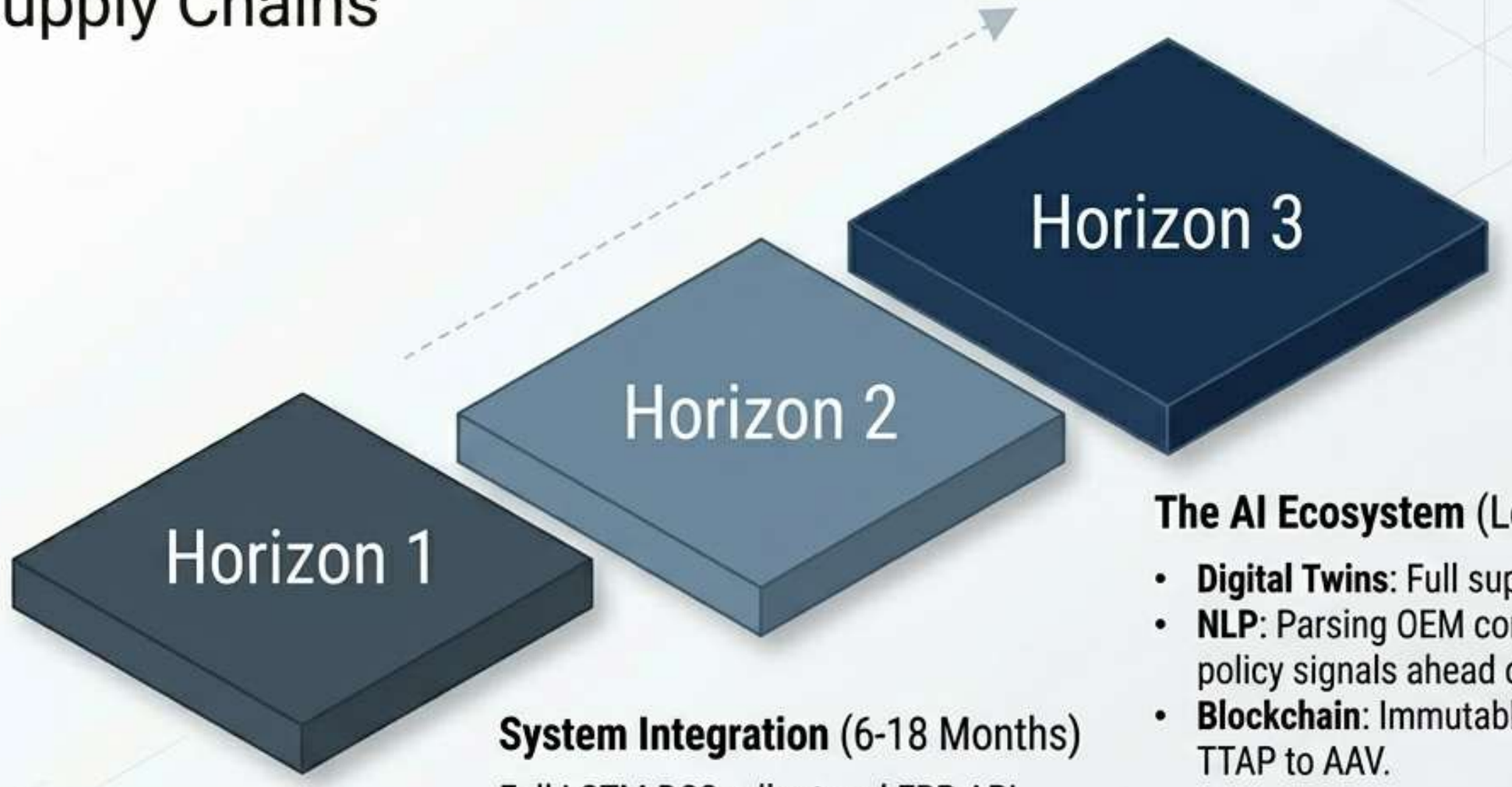
The Strategic Horizon for Cognitive Cognitive Supply Chains



Immediate Value (0-3 Months)

Deploy AMIC regression formula manually as a 120-day pre-lot checkpoint.

System Integration (6-18 Months)

Full LSTM-DSS rollout and ERP API integration for Toyota Fortuner; extend training data to GM and Nissan programs.

The AI Ecosystem (Long-term)

- **Digital Twins**: Full supply chain stress testing.
- **NLP**: Parsing OEM communications and policy signals ahead of AMIC data.
- **Blockchain**: Immutable lot tracking from TTAP to AAV.

Closing Takeaway: Transforming Mobica into the most market-aware, analytically rigorous automotive supplier in Egypt.

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Presented to

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